who currently access proprietary DOB feeds and would have switched to using consolidated market data under the adopted rule, with additional information on liquidity provision across more price levels. To the extent that these market participants can utilize full depth of book information, the Commission believes that this alternative could result in increased benefits to such market participants relative to the adopted rule.[2502] Certain commenters on the Roundtable stated that without full depth of book information, broker-dealers may not be able to provide best execution to their clients,[2503] indicating that full depth of book information would provide valuable information to market participants. However, as discussed above, the Commission believes that the marginal benefit of including additional information on price levels further away from the best quotes may decrease as the price level moves away from the best quote because orders at these price levels are less likely to execute.[2504]

Relative to the adopted rules, the inclusion of full depth of book information in core data would increase the ability of market participants to use it as a substitute for proprietary DOB feeds.[2505] Currently, market participants interested in full depth of book data rely on proprietary DOB feeds offered by exchanges, which provide varying degrees of the depth of book

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2502 This alternative could increase costs relative to the adopted rule for market participants that access full depth of book information and execute trading that earn profits at the expense of other market participants who do not access this information. As discussed above, this cost would represent a partial transfer from traders who currently have access to depth of book to those who do not. See supra Section V.C.1(c)(iv).

2503 See Proposing Release, 85 FR at n.284–85.

2504 See supra Section V.C.1(c)(ii).

2505 Including full depth of book information in core data would not make it a perfect substitute for all proprietary DOB feeds. For example, some proprietary DOB feeds contain more detailed information than full depth of information, such as messages on individual orders.

information. To the extent that there are market participants who utilize full depth of book information via proprietary DOB feeds in trading, this alternative could increase the benefits for some of these market participants relative to the adopted rules by potentially reducing their data costs if they would switch to using core data under this alternative but would not have done so under the adopted rules. Subscribers of proprietary DOB feeds would realize these cost savings if they switched to receiving consolidated market data through a competing consolidator product or if they registered as a self-aggregator.[2506]

The Commission believes that the alternative to include full depth of the book in core data would result in greater costs for exchanges than would the adopted rules. To the extent that the alternative results in fewer market participants subscribing to proprietary DOB data or purchasing connectivity services from the exchanges than under the adopted rules, exchanges' business for their proprietary feeds and connectivity services could be less profitable.[2507] Additionally, to the extent that not all exchanges sell full depth of book, certain exchanges would incur additional costs to set up systems and produce full depth of book information to be included in the core data. However, the Commission is unable to quantify this cost because it lacks information on the modifications exchanges would need to make to their systems in order to provide full depth of book information.

2506 See supra Section V.C.2(b).

2507 More broadly, this could have differential effects between exchanges who derive significant revenue from proprietary data feeds and those who derive significant revenue primarily from SIP revenue. These effects would also depend on the effective national market system plan(s) fees for consolidated market data offerings as well as their method for allocating revenue received from consolidated market data among the SROs. See supra Section V.C.4(a).

Compared to the adopted amendments, this alternative could result in additional costs for competing consolidators to create infrastructure and expand capacity to distribute full depth of book information.[2508] The costs are likely to vary substantially according to the existing infrastructure of the entity seeking to be a competing consolidator. The Commission believes that these incremental costs for market data aggregators and existing exclusive SIPs will be small, because they already work with proprietary DOB data.

Additionally, including full depth of book information would require market participants who subscribed to core data and wished to receive the additional depth of book information to make more extensive upgrades to their systems than under the adopted rules. However, the Commission is unable to estimate the associated costs because it does not have access to information about the infrastructure expenses a market participant incurs to process market data and because of the likelihood that such costs vary substantially according to the existing infrastructure of the market participant. To the extent that some market participants who subscribe to the exclusive SIPs do not need full depth of book information, they would not need to expand their own proprietary technology or that of a third-party vendor to process the full depth of the book data. Therefore, this alternative would not result in additional costs for these market participants compared to the adopted rules.

2. Introduce Changes in Core Data and Introduce a Distributed SIP Model

The Commission considered an alternative that would expand the core data as proposed and would introduce a distributed SIP model whereby the current exclusive SIP processors

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[2508] See supra Section V.C.2(d).

would establish multiple instances of their systems in multiple data centers.[2509] As the Roundtable panelists[2510] stated this alternative would achieve a similar reduction in exclusive SIP geographic latency to the adopted rule by allowing firms to consume data under the current structure without making any changes or to consume data at the nearest exclusive SIP instance depending on the firms' latency concerns.[2511] However, this alternative would still provide exclusive rights to one operator to provide exclusive SIP services for a given tape.

The Commission believes that this alternative would produce lower benefits compared to the decentralized consolidation model.[2512] The Commission believes that under this alternative, the exclusive SIPs would not be subject to the same competitive forces that competing consolidators may be subject to under the decentralized consolidation model.[2513] This lack of competition would reduce the incentives to innovate and would not improve efficiency or reduce the transmission and aggregation latencies of core data as much as the proposal. If core data does not achieve the same overall latency reduction as under the adopted rule, then market participants would be less likely to substitute using core data for proprietary data than they

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2509 See also Proposing Release, 85 FR at Section IV.C.2 for a discussion about a single SIP alternative.

2510 See Proposing Release, 85 FR at Section IV.C.1(a).

2511 Several commenters agreed. See, e.g., NYSE Letter II at 26; Nasdaq IV Letter at 36, 49; Cboe Letter at 25 for a discussion on the advantages of the distributed SIP alternative and how the commenters believe the Commission did not properly consider it.

2512 See supra Section V.C.2(c). See also Proposing Release, 85 FR at Section IV.C.1.

2513 One commenter agreed. See MEMX Letter at 8. See also supra Sections V.C.2, V.D.2.

would be under the adopted rule. This could mean that the potential decline in profits from exchanges' proprietary data fees may not be as large as they would be under the proposal.[2514]

Under this alternative, the exclusive SIPs would still need to make upgrades to their systems to account for the expansion of core data and would still need to install systems in multiple data centers. The Commission believes that the costs of these SIP system upgrades would be similar to those under the adopted rule for the exclusive SIPs that registers to become a competing consolidator.[2515] However, under this alternative, market participants may experience higher costs to access consolidated market data compared to the adopted rule. Instead of having the option to receive all consolidated market data from one competing consolidator, as they would under the adopted rule, market participants would still need to receive data from both exclusive SIP plan processors.[2516] This means that under this alternative, the total price market participants would pay to access consolidated market data may be greater than under the adopted rule because it would include the costs of the two plan processors to aggregate and transmit the data. Under the adopted rule, the total price market participants would pay to receive consolidated market data may only include the costs of one processor, because market participants would have the option to receive all of their consolidated market data products from one competing consolidator.[2517]

2514 See supra Section V.C.4(a).

2515 See supra Section V.C.2(d).

2516 See supra Section V.B.2.

2517 See supra Section V.C.2(c).

One commenter stated that the Commission relies on a limited set of information when examining potential solutions to the latency differential between the proprietary and SIP data feeds and "does not consider any other approaches to resolving its latency concerns."[2518] Commenters also emphasized that a distributed SIP alternative would introduce much less regulatory disruption and would create a more resilient market than the one with competing consolidators.[2519] The Commission disagrees with the first comment because the Commission considered a Distributed SIP alternative in its Proposing Release. However, as mentioned above, the Commission believes this alternative would produce lower benefits compared to the decentralized consolidation model because it lacks the competitive incentives achieved in the decentralized consolidation model. Additionally, the Commission disagrees with the second comment and believes that the decentralized consolidation model will increase market resiliency, as discussed above.[2520]

3. Require Competing Consolidators' Fees be Subject to the Commission's Approval

The Commission considered an alternative to the decentralized consolidation model that would require competing consolidators' fees to be subject to the Commission's regulatory approval. Some commenters supported this alternative[2521] and one commenter stated its recommendation "that the Commission should scrutinize competing consolidator fees, and fee

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2518 See Nasdaq Letter IV at 45.

2519 See, e.g., Cboe Letter at 25; Nasdaq Letter IV at 36; Data Boiler Letter I at 66–67.

2520 See supra Section V.C.2(c)(iv) for a discussion on the resiliency benefits of the decentralized consolidation model.

2521 See, e.g., Clearpool Letter at 4; ACS Execution Services Letter at 5–6.

changes, in a manner similar to the process for review and approval of proposed rule changes currently filed by SROs."[2522]

The Commission believes that, relative to the adopted rule, this alternative would potentially reduce the risk and uncertainty surrounding the total price of consolidated market data. This alternative would provide for Commission review and approval of the fees of competing consolidators. Therefore, compared to the amendments, this alternative could reduce the risk that market participants are exposed to unreasonably high fees, which could reduce the risk that some market participants or data vendors would no longer provide services in the equity market because the price of consolidated market data products becomes too high.[2523]

The Commission believes, however, that this alternative would impose additional regulatory burdens on the competing consolidator business compared to the adopted rule, and may inhibit competing consolidators from being able to respond effectively and quickly to free market forces. These burdens would reduce the incentive for firms to become competing consolidators and lead to less robust competition in the decentralized consolidation model than under the adopted rule.[2524] With less competitive forces to discipline competing consolidators' service fees, competing consolidators would have less incentive to innovate in their consolidating business. Moreover, less competing consolidators in the market would reduce the extent to which the pricing is based on market forces. Finally, the Commission believes that under the amendments, there will be a competitive market for consolidated market data products with several competing consolidators operating. Thus, competitive forces will constrain the

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2522 See Clearpool Letter at 4.

2523 See supra Section V.C.2(b).

2524 See supra Section V.C.2(a).

prices competing consolidators can charge without the need to impose additional regulatory burdens on the competing consolidator business.

4. Do Not Extend Regulation SCI to Include Competing Consolidators

The Commission considered an alternative that would not extend Regulation SCI to include SCI competing consolidators.[2525] Under this alternative, there would be no SCI competing consolidators and the Commission would have required all competing consolidators to be subject to the provisions of Rule 614(d)(9).[2526] The Commission believes that this alternative would reduce some of the benefits as well as some of the costs compared to the adopted rules.[2527]

The Commission believes that this alternative could result in larger competing consolidators, that would have met the threshold for SCI competing consolidators under the

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[2525] One commenter preferred a similar alternative to the proposed requirement that all competing consolidators be SCI entities. This commenter believed that even if a firm's systems met the standards of Regulation SCI, demonstrating compliance with Regulation SCI would be costly and overly burdensome and act as a barrier to entry for firms seeking to become competing consolidators. See ACTIV Financial Letter at 2. As discussed above, the Commission agrees that the costs of Regulation SCI would serve as a barrier to entry to new competing consolidators. As discussed above, the Commission has adopted Rule 614(d)(9) that will apply to all competing consolidators during the initial one year transition period and competing consolidators below a threshold thereafter. The Commission believes that Rule 614(d)(9) will be less costly than Regulation SCI and will lower the barriers to entry for new competing consolidators. See supra Section III.F and Section V.C.2(e)(i).

[2526] Under this alternative, the Commission would also exempt competing consolidators affiliated with exchanges from the requirements of Regulation SCI if they complied with the provisions of Rule 614(d)(9), so they would not face higher regulatory burdens and be placed at a competitive disadvantage relative to competing consolidators that are not affiliated with exchanges.

[2527] See supra Section V.C.2(e)(i).

adopted rules, producing systems that would be less secure and resilient than they would be under the adopted rules because they would not be subject to all of the requirements of being an SCI competing consolidator.[2528] If these competing consolidators produce less secure and resilient systems compared to if they were SCI competing consolidators, then there could be a greater risk of more market disruptions due to systems issues in competing consolidators compared to the adopted rules.[2529] Additionally, if one of these competing consolidators does experience a systems issue, it could result in more severe and longer disruptions compared to the adopted rules. However, the increase in competing consolidator systems issues compared to the adopted rules may not be significant. Under this alternative, competing consolidators would still be subject to the requirements of Rule 614(d)(9) and would need to establish policies and procedures to ensure that their systems have levels of capacity, integrity, resiliency, availability, and security adequate to maintain operational capability. They would also still need to post information on systems issues on their websites as well as monthly reports containing statistics on their capacity and systems availability.[2530] This would place competitive pressure on competing consolidators to ensure that their systems are reliable and resilient. Otherwise, they could lose subscribers to competing consolidators that had more reliable and resilient systems.

The Commission believes that this alternative would result in lower costs for larger competing consolidators compared to the adopted rule. Under this alternative, these competing

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2528 For example, under this alternative, larger competing consolidators would not have the requirements to have geographically diverse back-up and recovery capabilities. See supra Sections III.F and V.C.2(e)(i).

2529 Id.

2530 See supra Section V.C.3(a).

consolidators would not incur the costs that are associated with being an SCI competing consolidator that are discussed above.[2531] Instead, these competing consolidators would have to bear the lower costs associated with Rule 614(d)(9).[2532]

The Commission believes that these lower costs could result in more firms becoming competing consolidators and could increase competition in the competing consolidator market compared to the adopted rules. Although, under the adopted rules, competing consolidators will initially be subject to the lower costs of Rule 614(d)(9), the Commission believes that many competing consolidators will eventually meet the market data revenue threshold for SCI competing consolidators and be subject to the higher costs associated with Regulation SCI. The lower costs under this alternative may result in more firms becoming competing consolidators compared to the adopted rules, which could increase competition. An increase in competition may increase the benefits from the decentralized consolidation model. However, these effects may not be significant because the Commission believes that the risk under the adopted rules that the anticipated benefits of the decentralized consolidation model will not materialize because of insufficient competition among competing consolidators is low.[2533] To the extent these effects do occur, it could lower the costs and increase the speed and quality of consolidated market data products compared to the adopted rule. This, in turn, could make consolidated market data products a more viable substitute for proprietary data feeds and result in greater competition between consolidated market data and proprietary data feeds compared to the adopted rules.

2531 See supra Section V.C.2(e)(ii).

2532 See id.

2533 See supra Section V.C.2(a)(ii).